

Deep Learning Systems Analysis Report

Github Repo Link: <https://github.com/Shehzaad-S/deep-learning-systems-project>

Report Overview

This project addressed a conversational language modeling task using the DailyDialog dataset, a manually curated collection of multi-turn human dialogues designed to support natural language processing research (Li et al., 2017). A GPT-style decoder-only Transformer model was implemented from first principles using PyTorch and trained to perform autoregressive next-token prediction, following the Transformer architecture introduced by Vaswani et al. (2017) and the generative pre-training paradigm proposed by Radford et al. (2018). A baseline architecture was evaluated alongside controlled depth and width experiments to investigate how architectural modifications influence conversational language modeling performance, parameter efficiency, and generalization ability.

Dataset and Task Description

The DailyDialog dataset is a manually curated collection of multi-turn human conversations designed to support dialogue system and natural language processing research (Li et al., 2017). The corpus contains 11,118 conversations partitioned into predefined training, validation, and testing splits, covering a wide range of everyday topics including relationships, education, work, culture, health, finance, and daily life. Unlike many large-scale conversational datasets collected from online forums or social media, DailyDialog was constructed to reflect coherent human interactions and realistic conversational patterns, making it well suited for conversational language modeling tasks.

For this project, the dataset was used to train an autoregressive language model that predicts the next token in a sequence given all preceding tokens. The original dialogue files were preprocessed by preserving conversation boundary markers (`__eou__`) and appending end-of-sequence tokens to each conversation.

Dataset Link: <https://huggingface.co/datasets/roskoN/dailydialog>

Model Architecture and Design Decisions

The baseline model followed a GPT-style decoder-only Transformer architecture built using the self-attention framework introduced by Vaswani et al. (2017) and the autoregressive language modeling approach popularized by GPT models (Radford et al., 2018). A

pretrained GPT-2 tokenizer was selected because it provides an established subword vocabulary capable of efficiently representing conversational text while reducing out-of-vocabulary issues commonly encountered with word-level tokenization approaches. End-of-sequence tokens were preserved throughout preprocessing to allow the model to learn conversational turn-taking behavior and dialogue structure directly from the training data.

The baseline architecture consisted of token and positional embeddings, two Transformer blocks, four attention heads, a hidden embedding dimension of 128, and a feed-forward dimension of 512. Multi-head causal self-attention was implemented to ensure that each token could only attend to preceding tokens, preserving the autoregressive next-token prediction objective described by Vaswani et al. (2017). A context length of 256 tokens was selected to balance conversational context coverage against computational cost, while the AdamW optimizer was used for training due to its decoupled weight decay formulation and demonstrated effectiveness in deep learning optimization (Loshchilov & Hutter, 2019).

The architectural configuration was intentionally designed to remain computationally feasible within available hardware constraints (colab gpu) while still providing sufficient capacity to learn meaningful conversational language patterns. This baseline configuration also provided an appropriate foundation for subsequent architectural experiments involving increases in Transformer depth and embedding dimensionality. We did not re-invent the wheel here, we used proven design principles, I truly commend the amazing researchers who developed these model architectures.

Experimental Comparison

To evaluate how architectural design choices influence conversational language modeling performance, two controlled experiments were conducted in addition to the baseline model. The first experiment increased Transformer depth by doubling the number of Transformer blocks from two to four while keeping all other architectural and training parameters constant. This modification was selected because deeper Transformer models can perform additional rounds of contextual processing and representation learning through successive attention and feed-forward operations, potentially improving language modeling performance (Vaswani et al., 2017).

The second experiment investigated model width by increasing the embedding dimension from 128 to 256 while preserving the original two-layer architecture. To maintain the standard Transformer design ratio, the feed-forward network dimension was increased proportionally from 512 to 1024. This experiment was selected to evaluate whether **richer token representations** and **increased model capacity** could improve conversational language modeling performance.

Both experiments were designed as controlled comparisons in which only a single major architectural characteristic was modified at a time. All other aspects of the training process, including the dataset, tokenizer, context length, batch size, learning rate, optimizer, dropout rate, and number of training epochs, were held constant. This approach ensured that observed differences in performance could be attributed primarily to the architectural changes under investigation rather than variations in the training configuration.

Results and Interpretation

The baseline GPT-style decoder-only Transformer achieved a final training loss of **2.7638**, a validation loss of **3.1903**, and a test loss of **3.1156**. These results demonstrated that the model successfully learned conversational patterns from the DailyDialog corpus and provided a strong foundation for controlled architectural experimentation.

The first experiment increased Transformer depth from two to four layers while maintaining the original embedding dimension and training configuration. This modification reduced the validation loss from **3.1903** to **3.0442** and produced the best overall test loss of **2.9701**. Notably, these improvements were achieved while increasing the parameter count from approximately **13.3 million** to only **13.7 million** parameters. The results suggest that additional Transformer blocks enabled richer contextual processing and improved generalization to previously unseen conversations.

The second experiment increased the embedding dimension from 128 to 256 while preserving the original two-layer architecture. This wider model contained approximately **27.4 million** parameters and achieved the lowest training loss (**2.1817**) and a reduced test loss (**3.0356**) relative to the baseline. However, the improvement in validation and test performance was smaller than that observed in the depth experiment despite the substantially larger increase in model size.

Overall, both architectural modifications improved performance compared to the baseline model, confirming that increased model capacity can benefit conversational language modeling. However, the depth experiment produced the most favorable trade-off between performance and computational efficiency, achieving the strongest generalization performance while requiring only a modest increase in parameter count. In contrast, the width experiment demonstrated that larger embedding spaces improved the model's ability to fit the training data but yielded diminishing returns on unseen data relative to the additional computational cost.

Limitations and Risks

Several limitations should be considered when interpreting the results of this study. First, although the DailyDialog dataset provides high-quality conversational data, it represents a relatively small corpus compared to the large-scale datasets commonly used to train modern language models, potentially limiting the diversity of linguistic patterns and conversational behaviors available during training. As a result, generated responses occasionally exhibited grammatical fluency while lacking deeper semantic consistency or factual coherence.

A second limitation relates to the computational constraints under which the experiments were conducted. Model architectures, context length, training duration, and experimental configurations were selected to remain feasible within the available hardware resources. Larger models, longer training schedules, or additional hyperparameter optimization may have produced further performance improvements but were beyond the scope of this project.

Finally, evaluation was primarily based on training, validation, and test loss measurements together with **qualitative inspection** of generated dialogue. While these metrics provide valuable insight into language modeling performance, they do not fully capture response quality, factual accuracy, coherence across long conversations, or user-perceived conversational effectiveness. More comprehensive evaluation procedures involving human assessment or additional language generation benchmarks would be beneficial in future work.

Ethical and Responsible Use

One ethical consideration associated with conversational language models is the potential for generating inaccurate, misleading, or biased responses. Because the model learns conversational patterns directly from its training data, any biases, omissions, or limitations present within the dataset may influence generated outputs. Additionally, despite learning conversational structure effectively, the model does not possess factual understanding and may produce responses that appear plausible while containing incorrect or nonsensical information.

From a responsible AI perspective, systems based on conversational language models should therefore be deployed with appropriate safeguards, transparency regarding their limitations, and human oversight when used in settings where inaccurate information could negatively affect users. These considerations become increasingly important as language models are applied to domains such as education, healthcare, customer service, or decision support along with any other decision-based faculties that rely on the models generation capabilities.

Future Improvements

Several realistic improvements could be explored with additional time, computational resources, and training data. The experimental results suggest that increasing Transformer depth was a particularly effective strategy for improving generalization performance, indicating that future architectures could combine the stronger aspects of both depth and width modifications to better balance model capacity and parameter efficiency. Additional experiments involving larger context lengths, alternative attention configurations, regularization strategies, and extended training schedules may further improve conversational language modeling performance.

From a data and representation perspective, an interesting future direction would be the development of a custom tokenizer tailored specifically to the DailyDialog corpus. Exploratory analysis showed that only a portion of the GPT-2 vocabulary was utilized during training, suggesting that a smaller domain-specific vocabulary could potentially improve token efficiency and reduce model complexity while preserving conversational coverage. Investigating tokenizer design and vocabulary optimization may therefore provide another avenue for improving performance beyond architectural scaling alone.

More comprehensive evaluation procedures, including human judgment studies and additional language generation benchmarks, would also provide a more complete assessment of conversational quality than loss-based metrics alone. Finally, access to greater computational resources would support larger-scale architectural experimentation and potentially enable training regimes more comparable to those used in modern large language models.

References

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